

Multiple Choice (10 marks)

1. Short sequence by promoter that assists transcription by interacting with regulatory proteins.
 - (a) Promoter
 - (b) Inducer
 - (c) Repressor
 - (d) Operator
 - (e) Inhibitor
2. Vinegar is classified as an acid. What does this tell you about the concentration of the hydrogen ions and the hydroxide ions?
 - (a) Vinegar has a lower concentration of hydrogen ions than hydroxide ions
 - (b) Vinegar has an excess of hydrogen ions with respect to hydroxide ions
 - (c) Vinegar has a higher concentration of hydroxide ions than water
 - (d) Vinegar has no hydroxide ions
 - (e) Vinegar does not contain hydrogen ions
3. Why does milk "when kept in a refrigerator?
 - (a) Lactose crystallization at low temperatures causes milk to spoil
 - (b) Milk spoils due to absorption of odors in the fridge
 - (c) Milk spoils due to the presence of mesophiles
 - (d) Milk spoils due to chemical reactions in the fridge
 - (e) Thermophiles cause milk to spoil in the refrigerator
4. In a commensalistic relationship between two species living symbiotically what are the consequences of their interactions?
 - (a) Both species benefit equally
 - (b) One species benefits, and the other is significantly harmed
 - (c) One species harms the other without any benefit
 - (d) Both species harm each other
 - (e) The species exchange benefits, but one benefits more than the other
5. What is the evidence for the theory that the neurilemma sheath plays a role in the regeneration of severed nerves?

- (a) The neurilemma sheath absorbs toxins at the injury site to facilitate regeneration.
- (b) The neurilemma sheath provides a channel for the axon to grow back to its former position.
- (c) The neurilemma sheath fuses severed axon ends together.
- (d) The neurilemma sheath repels inhibitory molecules that prevent nerve growth.
- (e) The neurilemma sheath acts as a source of neural stem cells for regeneration.

True / False (10 marks)

Answer True or False

- 6. True or False: Silk worms spin cocoons based on the phase of the moon, indicating a link between lunar cycles and hormonal regulation.
- 7. True or False: A DNA molecule with 180 base pairs and 20 percent adenine contains 72 cytosine nucleotides.
- 8. True or False: The typical periods in the unfolding of a behavioral act include appetitive behavior, consummatory act, and quiescence.
- 9. True or False: Radial cleavage involves cell divisions that occur in a non-oblique manner, resulting in symmetrical arrangements around the polar axis.
- 10. True or False: Disruptive selection occurs when intermediate traits are favored, resulting in increased variance in birth weights among human babies.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. Discuss the concept of stabilizing selection in the context of Swiss starlings, focusing on how clutch size affects fitness and evolutionary outcomes.
- 12. Discuss the key characteristics that differentiate prokaryotic cells from animal cells, focusing on structural components such as the cell wall and their implications for cellular function.

End of Paper